Reg. No.									

MANIPAL INSTITUTE OF TECHNOLOGY

MANIPAL

(A constituent unit of MAHE, Manipal)

THIRD SEMESTER B.TECH. (ELECTRICAL &ELECTRONICS)

END SEMESTER EXAMINATIONS, NOVEMBER- DECEMBER 2024

MICROCONTROLLERS [ELE 2125]

REVISED CREDIT SYSTEM

Time: 3 Hours **Date: 20 NOVEMBER 2024** **Max. Marks: 50**

Instructions to Candidates:

- Answer **ALL** the questions.
- Missing data may be suitably assumed.

1A. Outline the address of Bit addressable RAM locations. How is the bit addressing locations are distinguished from byte wise addressing by 8051 microcontrollers. Explain with an example. ***(03)***

1B. Define subroutine? Illustrate the instructions involved in calling a subroutine with an example. ***(03)***

1C. Write an 8051-assembly language program to add ten '8' bit BCD numbers available at successive external RAM memory locations starting at 5000H and store the 16-bit BCD result in 6000H and 6001H. ***(04)***

2A. Write 8051 microcontroller alternate assembly language instruction/s to perform the same operation as (i) PUSH 0F0H (ii) INC DPTR. ***(02)***

2B. Develop an 8051-assembly language program to count the number of people crossing an entrance. Assume that each time a person crosses the entrance, a pulse is generated and is applied to T1/P3.5 pin of 8051. After every second, load the count value to port '2' and port '3', interfaced to two separate LCD units for display. Reset the count back to 0000H, after a maximum value of FFFH is reached. Use timer 0 in mode 2 to obtain the required delay of one second. ***(04)***

2C. Assume that 3 pins of 8051 P1.0, P1.1, and P1.2 are connected to 3 LEDs (Red, Orange, Green) of traffic light signals. The green light glows for 35 seconds, followed by the orange light for 5 seconds and the red light for 40 seconds during business hours (8 AM–8 PM). At night (8 PM–8 AM), the green light glows for 25 seconds, followed by the orange light for 5 seconds and the red light for 30 seconds. Show the down counter time for each LED display in port "2" in addition to the LED light display, with a delay of 1 sec between each count. Use 8051 instructions to obtain the required delay. Assume that an interrupt $\overline{INT0}$(P3.2) is activated by an external device for every 12 hours. Given: XTAL=11.0592MHz. ***(04)***

3A. Explain the functions of the following signals of 8051 with respect to external memory interfacing.

(i) $\overline{PSEN}$ (ii). $\overline{EA}$ (iii) $\overline{RD}$ ***(02)***

3B. Interface a 4x3 matrix keyboard with 12 characters as shown in figure to 8051 microcontrollers. The characters in the keyboard are interfaced to 8051 using P1.0, P1.1, P1.2, P1.3(for rows) and P2.0, P2.1, P2.2(for columns) pins. Develop an interfacing diagram of 4x3 matrix keyboard to 8051 and write an 8051 ALP to detect any key pressed and transfer the ASCII code of key pressed to port '3'. Use software debouncing technique.



(04)

3C. A temperature sensor is interfaced with the 8051 through an 8- bit ADC connected on Port 1. Develop an assembly language program to transmit serially the message specified below at 19200 baud.

If ADC output is less than or equal to 30H, serially transmit the message - "LOW".

If ADC output is greater than 30H, serially transmit the message-"HIGH".

8051 is clocked at 11.0592MHz. (Hint: ADC output is available in port 1). ***(04)***

4A. Find the frequency and duty cycle of the rectangular waveform generated by the program code given below. Assume XTAL=16 MHz

Program:

```
        MOV TMOD, #20H
        MOV TH1, #46H
        SETB TR1
  HERE: SETB P1.5
        ACALL DELAY
        ACALL DELAY
        ACALL DELAY
        CLR P1.5
        ACALL DELAY
        SJMP HERE
DELAY: JNB TF1, DELAY
        CLR TF1
            RET
```

(03)

4B. The following C code was used to clear fifty-word locations in memory. Translate it to the ARM7 assembly program. Use suitable registers and memory locations.

```
for (i=0; i<50; i++)
        {
         a[i]=0;
                }
```

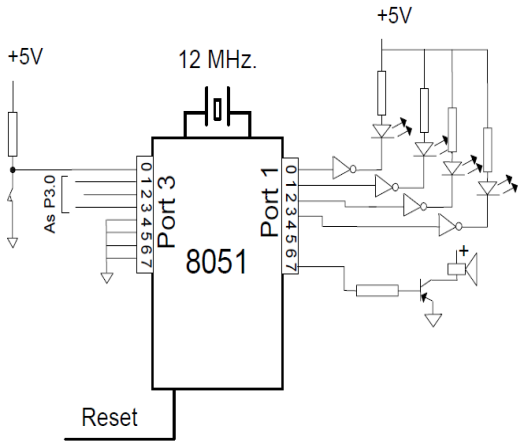
(03)

4C. Write an ARM ALP to multiply two 32-bit numbers available in memory locations 0xA000-0xA003 and 0xA004-0xA007. Add the 64-bit product to another 64-bit number stored in 0xA008-0xA00B (LS word) and 0xA00C-0xA00F (MS word). Store the 64-bit result in the next eight locations. ***(04)***

5A. Explain the architecture of 8051 microcontroller. ***(03)***

5B. Write ARM7 assembly language program to multiply the 32-bit number in r0 by the 115 without using multiplication instruction. ***(03)***

5C. A simple burglar alarm system has 4 zone inputs connected to pins P3.0 to P3.3 of 8051. If any one of these inputs is activated(low) a bell will sound for 5 minutes and the corresponding zone LED (P1.0 to P1.3 respectively), will be activated. Design an 8051-assembly language program to implement the required functionality for this system shown in Figure.



(04)